NETWORK COMMANDS

Exp no: 1

Aim:

To study various network commands used in linux and windows.

Procedure:

1. Open the command prompt / terminal.
2. Execute the required network commands
3. Observe and record the output of each command.
4. Analyse the network information displayed by the commands.

Commands:

1. APP command
   ~ arp -a
   Purpose: Display IP and MAC address information.
2. Hostname:
   ~ hostname
   Purpose: Display name of the computer.
3. IP configuration
   ~ ipconfigall
   Purpose: Display detailed TCP/IP configuration information.
4. Netbios
   ~ netstat -a
   Purpose: Displays network connects and routing info.
5. Network statistics:
   ~ Net stat -r
   Purpose: Displays NetBios name information.
6. DNS lookup:
   ~ nslookup www.google.com.
   Purpose: Display DNS information of the specified domain.

Sample Output:

C:\> arp -a

Interface : 192.168.1.10

Internet address	Physical Adress	Type
192.168.1.1	00-11-22-33-44-55	Dynamic

C:\> hostname

DESKTOP-PC

C:\> ipconfig/all

IPv4 Address: 192.168.1.10

Default Gateway: 192.168.1.1

DNS server : 8.8.8.8

C:\> net stat -r

Network destination	gateway
0.0.0.0	192.168.1.1

C:\> nslookup www.google.com

Name: www.google.com

Address : 142.250.193.68.

C:\> ping google.com

Reply from 142.250.195.68 bytes = 32 time = 20ms TTL = 117

Reply from 142.250.195.68 bytes = 32 time = 18ms TTL = 117

7. Ping:

~ ping google.com

Purpose: Checks connectivity between the local system and a destination host.

8. Route

~ route

Purpose: Displays or modifies the IP routing table.

9. IP command

~ ip address show

Purpose: Display IP address assigned to network interfaces.

10. MTR

~ mtr google.com

Purpose: Displays the route and network statistics for each hop.

11. TCP Dump

~ tcpdump -i eth0

Purpose: Captures and displays network packets.

12. NMCLI

~ nmcli connection show

Purpose: Displays network Manager connection profiles.

Result:-

Thus, the command to run network configuration IP address, MAC address, routing information, DNS details and connectivity status user displayed succesfully.

Sample output

Straight - through cable.

End A → T568B

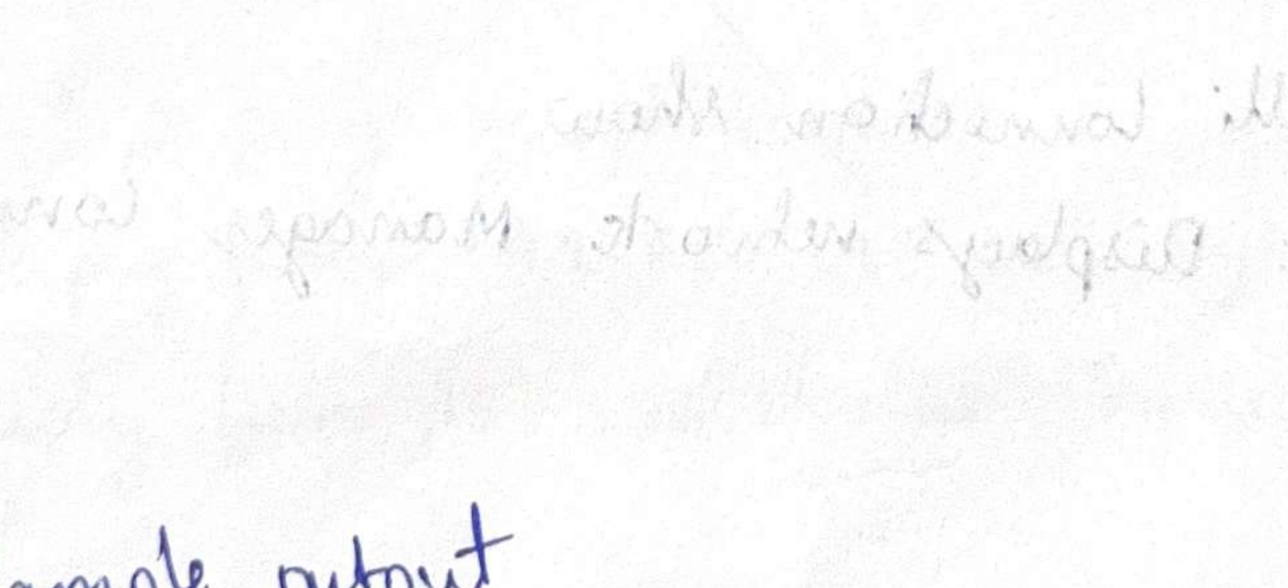
End B → T568B

Crossover cable:

End A → T568A

End B → T568B

cable tested successfully.



Exp no: 02 STUDY OF NETWORK CABLES

Aim:

To study the different types of network cables and to make straight - through and crossover ethernet cables using a crimping tool.

Procedure:

1. Take the required CAT3/CAT5e Ethernet cable, crimping tool and two RJ45 connectors.
2. Strip approximately 1.5cm of the outer insulation from both ends of the cable.
3. Untwist the four pairs of wires and arrange them in the required colour sequence
4. For a straight-through cable, use the same wiring arrangement at both ends. For a crossover cable, use arrangement A at one end and arrangement B at the corner.
5. Align the wires evenly and insert them carefully into RJ45 connector without changing their order.
6. Insert the RJ45 connector without changing into the crimping tool and crimp it firmly.
7. Repeat the procedure for the other end and test the cable by connecting the required network devices.

Result:

Thus, the different types of network cables were studied and the straight-through and crossover ethernet cables were successfully prepared and tested using RJ45 connectors and a crimping tool.

Sample output

Hub Network

PC0 → Hub → PC1

PC0 ↔ Hub → PC2

PC0 → Hub → PC3

Switch network:

PC0 → Switch → PC1

Hub forwards the packet to all connected ports.

CISCO PACKET TRACER

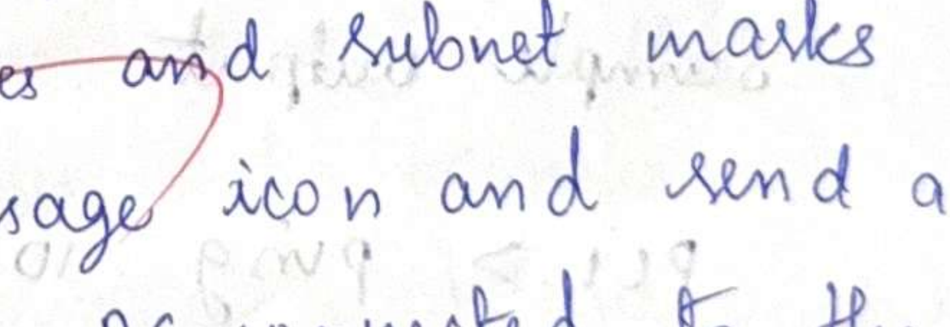
Exp no: 03

Aim:

To study the cisco packet tracer installation and user interface, understand the environment for designing a simple network, and analyse the behaviour of network device such as Hub and switch.

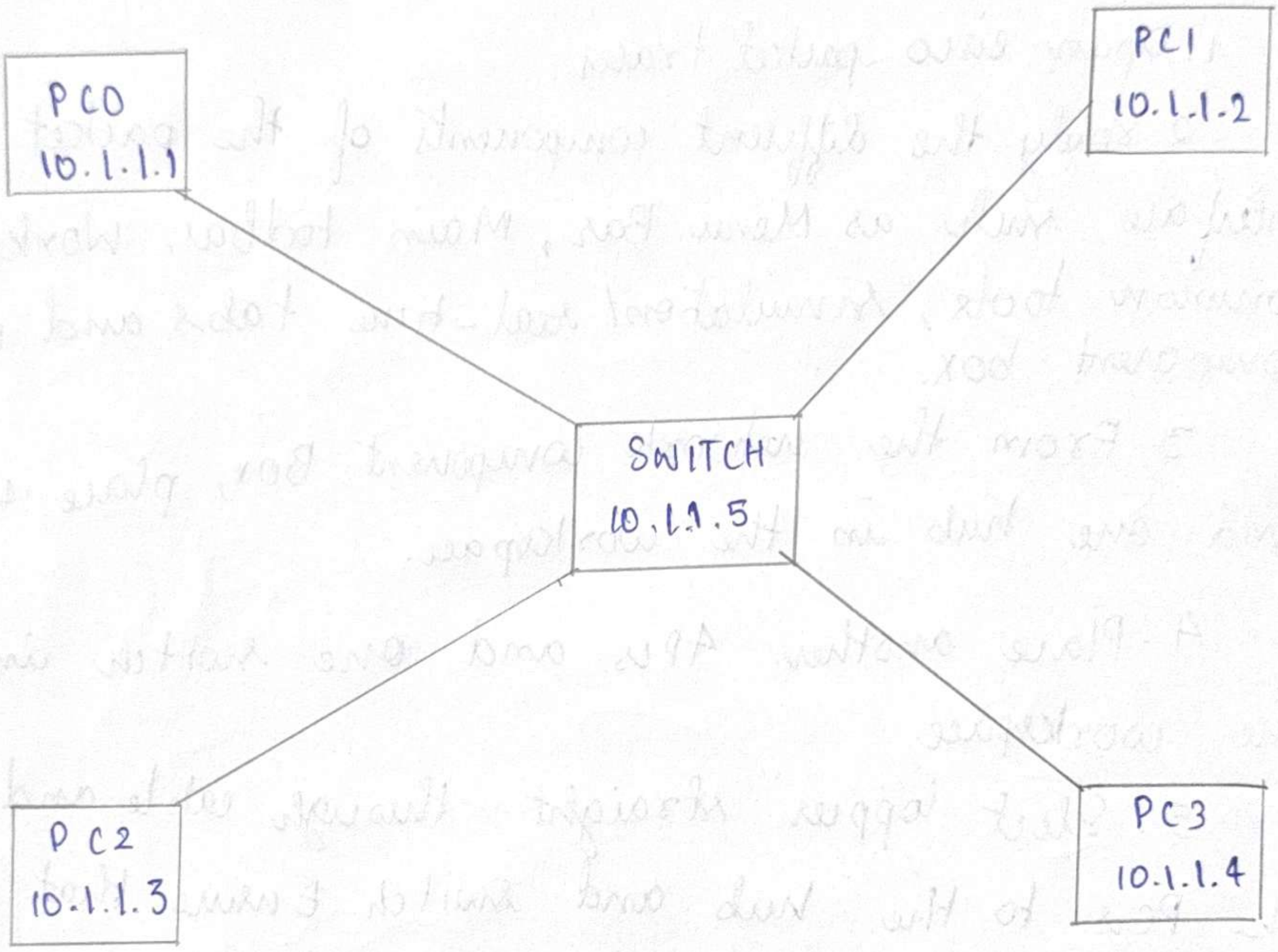
Procedure:

1. Open cisco packet tracer.
2. Study the different components of the packet tracer interface such as Menu Bar, Main toolbar, Workspace, common tools, simulation/ real-time tabs and network component box.
3. From the network component Box, place 4 PCs and one hub in the workspace.
4. Place another 4 PCs and one switch in the workspace.
5. Select copper straight-through cable and connect the PCs to the hub and switch. Ensure that the link LEDs turn green.
6. Open each PC and go to Desktop → IP configuration. Assign suitable IP addresses and subnet marks
7. Select the PDU/message icon and send a PDU from one PC to another PC connected to the Hub.
8. Observe the packet flow in the simulation.
9. Repeat the same process for PCs connected to the switch.
10. Observe and compare how the Hub & switch forward the packets.



Result:

Thus, the cisco packet tracer environment & user interface were studied, a simple network is designed.



Sample output

PC1 > ping 10.1.1.2

Reply from 10.1.1.2

Bytes = 32 Time <1ms TTL = 128

Packets : Sent = 4

Received = 4

Lost = 0

0% packet loss

SETTING UP A LAN

Exp no: 04

Aim:

To design, set up and configure a LAN connecting four computers through a network switch, assign static IP address, verify connectivity using the ping command and demonstrate data sharing.

Procedure:

1. Open Cisco Packet Tracer and place 4 PCs and one switch.
2. Connect all four PCs to the switch using copper straight-through cables.
3. Assign the following static IP addresses with subnet mask 255.0.0.0.
   - PC0 - 10.1.1.1
   - PC1 - 10.1.1.2
   - PC2 - 10.1.1.3
   - PC3 - 10.1.1.4
4. Open Desktop → Command prompt on a PC and using ping to check connectivity with the other PCs.
5. Add a server to the switch, assign an IP address and enable the HTTP service.
6. Access the server from another PC using the web browser to verify data transfer.
7. Use simulation mode to observe packet transmission through the switch.

Result:

Thus, a LAN with four computers and a switch was successfully configured and tested. Connectivity and data sharing between the devices were successfully verified.

Exp no:05

PACKET CAPTURE USING WIRESHARK

Aim:

To study the wireshark packet capture tool, capture network traffic and analyse different network packets and their protocol fields.

Procedure:

1. Install and launch wireshark on the computer.
2. select the required network interface from the capture section.
3. Double-click the interface to start capturing packets. The packets sent and received by the system will appear in real time.
4. Generate some network traffic and observe the captured packets.
5. Use the Packet list pane to view all the captured packets.
6. select any packet and use the Packet details pane to examine its protocols and protocol fields.
7. Observe the raw data of the selected packet in the Packet Bytes pane.
8. Observe the different colour codes used by wireshark to identify different types of network traffic.
9. Apply suitable display filters to identify and analyse specific network traffic.

Sample output

No.	Protocol	Source	Destination
1	ARP	192.168.1.10	Broadcast
2	DNS	192.168.1.10	8.8.8.8
3	TCP	192.168.1.10	142.xx.xx.xx
4	HTTP	192.168.1.10	Server.

10. Click the stop button after capturing the required packets.

11. Save the captured packets using File → Save for further analysis.

Result:

Thus, the network packets were successfully captured and analysed using Wireshark. The different network protocols and their corresponding packet fields were studied.

Sample output

Enter text : HELLO

Binary data:

0100100 01000101 01001100 01001100 0100111

Hamming code:

...

Error detected at position : 12

Error corrected successfully.

Decoded Data:

HELLO.

Exp no: 06

HAMMING CODE

Aim:

To implement error detection and correction using hamming code and verify the error correction capability using a test data stream.

Procedure:

Sender:

1. create a sender program
2. Read a text message from the input file.
3. Convert the text into binary.
4. Calculate and insert the required redundant parity bits.
5. Generate the Hamming code.
6. Store the encoded data in a file named channel.

Receiver

7. Read the encoded data from channel.
8. Recalculate the parity bits.
9. Determine whether an error is present.
10. If an error is detected, find its position.
11. Correct the erroneous bit.
12. Remove the redundant bits.
13. Convert the corrected binary data back into ASCII/text.

Result:

Thus, the Hamming code was successfully implemented to detect and correct a single-bit error in the transmitted data.

Exp no: 7

SLIDING WINDOW PROTOCOL

Aim:

To implement flow control at the Data link layer using sliding window protocol and simulate the transfer of frames between two nodes.

Procedure

Sender

1. Get the window size from the user.
2. Read a text message
3. Consider one character as one frame.
4. Assign a frame number to each character.
5. Create frames in the format. [Frame No._DATA]
6. Store the frames in Sender_buffer.
7. Send frames according to the window size.
8. Wait for acknowledgement from the receiver.
9. Read Receiver_buffer.
10. Check the ACK number.
11. If the ACK is correct, transmit the next set of frames.
12. If Nack is received, retransmit the required frame(s).
13. Introduce an error by manually modifying a frame/ACK number and observe the behaviour.

Receiver.

14. Read frames from Sender_buffer.
15. Check the frame number
16. If correct, generate ACK.

Sample output

Enter Window size : 3

Enter Message : HELLO

sender buffer:

Frame 0 : H

Frame 1 : E

Frame 2 : L

Receiver :

ACK 0

-ACK 1

ACK 2

Sender:

Frame 3 : L

Frame 4 : 0

Receiver :

ACK 3

ACK 4

Transmission successful.

With error

Frame 2 : Error detected

NACK 2 received

Frame 2 retransmitted

ACK 2 received.

17. If incorrect, generate NACK.

18. Store the response in receiver_buffer.

Result:

Thus, sliding Window Protocol was successfully implemented and frame flow control using ACK/NACK was verified.

Sample output

Starting Nmap scan...

Nmap scan report for 10.10.210.5

Host is up.

Nmap scan report for 10.10.210.6

Host is up.

Nmap done:

2 hosts up

Exp no: 8

NMAP LIVE HOST DISCOVERY

Aim:

To discover live hosts using Nmap scans based on ARP, ICMP, TCP and UDP on the TryHackMe: Nmap platform.

Procedure.

1. Open the Nmap room on TryHackMe.
2. Study the target network and subnet structure.
3. Identify the target IP addresses.
4. Perform ARP-based host discovery.
5. Use. Nmap -PR -sn TARGET
6. Perform ICMP echo discovery using: nmap -PE -sn TARGET.
7. Perform ICMP timestamp discovery using: nmap -PP -sn TARGET.
8. Perform TCP SYN Ping using: nmap -PS 23 TARGET
9. Perform TCP ACK-Ping using the appropriate -PA option.
10. Study UDP-based host discovery.
11. Observe the live hosts identified by Nmap.
12. Study reverse DNS lookup using -R.

Result:

Thus, Nmap was successfully used to discover live hosts using ARP, ICMP, TCP and UDP-based techniques.

Sample output

```
PC1 > ping 192.168.2.11

Reply from 192.168.2.11:
Bytes =32 Time<1ms TTL=126

Packets: Sent = 4
Received = 4
Lost = 0
0% Packet loss
```

Exp no: 9

SUBNETTING USING CISCO PACKET TRACER

Aim:

To implement classless IP subnetting using Cisco Packet Tracer and establish communication between multiple subnets through routers.

Procedure:

1. Open Cisco Packet Tracer.
2. Create a new network topology.
3. Add routers, switches and PCs.
4. Connect the devices using appropiate cables
5. Divide the network 192.168.1.0/24 into smaller subnets.
6. Use /27 subnet masks as specified in the manual.
7. Configure the router interfaces with suitable IP addresses.
8. Configure the switch ports.
9. Assign IP addresses and subnet masks to the PCs.
10. Configure default gateways.
11. Enable router interfaces using : no shutdown.
12. Test connectivity using ping.
13. Observe packet transmission between subnets.

Result:

Thus, classless IP subnetting was successfully implemented in Cisco Packet Tracer and communication between the subnets was verified.

INTERNETWORKING WITH ROUTERS

Exp no: 10

Aim:

To design and configure an interwork using routers in Cisco Packet Tracer and verify communication between different networks.

Procedure:

1. Open Cisco Packet Tracer.
2. Place one router and two PCs.
3. Connect the PCs to the router using copper straight-through cables.
4. Configure Router1.
5. Enter privileged mode: enable.
6. Enter configuration mode: config t
7. Configure FastEthernet 0/0:

```
interface FastEthernet 0/0.
ip address 192.168.10.1 255.255.255.0
no shutdown
```

8. Configure FastEthernet 0/1:

```
interface FastEthernet 0/1.
ip address 192.168.20.1 255.255.255.0
no shutdown
```

9. Configure PC0:

IP address : 192.168.10.2
subnet mask : 255.255.255.0
Gateway : 192.168.10.1

10. Configure PC1:

IP address : 192.168.20.2
subnet Mask : 255.255.255.0
Gateway : 192.168.20.1

Sample output

PC0 → Router1 → PC1

PDU status : Successful

PC1 → Router1 → PC0

Acknowledgement received.

11. Send a PDU from PC0 to PC1.

12. Observe the packet transmission and acknowledgement.

Result:

Thus, an internetwork using a router was successfully designed and configured, and communication between different networks was verified.

Sample output

Router # show ip route

R 192.168.20.0/24 [120/1]

via 192.168.10.2

Ping successful.

Packets : sent = 4

Received = 4

Lost = 0

STATIC ROUTING AND RIP

Exp no : 11

Aim :

To simulate static Routing and RIR (Routing information Protocol) using Cisco Packit Tracer.

Procedure

Part A - Static Routing

1. Open Cisco Packet Tracer
2. Place two routes and suitables PCs.
3. Connect the routers and PCs.
4. Assign IP addresses to all interfaces.
5. Configure router interfaces using ip address and no shutdown.
6. Add static routes using :

ip route <destination- network> <subnet -mask>

<next -hop>

7. Verify the routing table using :

show ip route.

8. Test connectivity using ping.
9. Configure the router interfaces.
10. Enter RIP configuration mode : router rip

version 2.

11. Add the connected networks. network <network-address>

12. Repeat the configuration on the other router
13. Verify the routing table.
14. Ping a host in the remote network.

Result:

Thus, Static Routing and RIP were successfully configured and the connectivity between different networks was verified.

Sample output

Server:
Server started
Waiting for client
client connected.

Received: Hello server
Echo: Hello server

Client:
Connected to server
Enter message: Hello server
Server replied:
Hello server

TCP/UDP socket COMMUNICATION

Exp no: 12

Aim:

To implement end-to-end communication at the Transport layer using TCP/UDP sockets and develop a simple chat application.

Procedure:

Echo client-server

1. Create a server program
2. Create a client program
3. Create a socket at the server.
4. Bind the socket to an IP address and port.
5. Make the server listen for client connections.
6. Accept the client connection
7. The client sends a message.
8. The server receives and echoes the message.
9. Display the response at the client.

chat Programs

10. Create client and server socket programs
11. Establish a connection
12. Reaad messages from the user.
13. Send the messages from the user.
14. Receive and display messages at the other end.
15. Continue communication until the user terminates the connection.

Result:

Thus, end-to-end communication was successfully implemented using socket programming and echo/chat communication between client and server.

Sample output:

Enter destination : 8.8.8.8

PING 8.8.8.8

Reply from 8.8.8.8
sequence = 1 Time = 24 ms

Reply from 8.8.8.8
sequence = 2 Time = 23 ms

Reply from 8.8.8.8
seqnence = 3 time = 25 ms

Packet sent : 3

Packet Received : 3

Packet Loss : 0 %

OWN PING PROGRAM

Exp no: 13

Aim:

To implement a simple Ping program using socket programming and verify the reachability of a destination host.

Procedure:

1. Create a program to accept a destination IP address / hostname.
2. Create a socket for network communication.
3. Generate an ICMP echo request.
4. Send the request to the destination.
5. Wait for an ICMP echo Reply.
6. Calculate the response time.
7. Display whether the host is reachable
8. Repeat the process for multiple requests
9. Display packet transmission statistics.

Result:

Thus, a simple Ping program was successfully implemented and the reachability of the destination host was verified.

Sample output:

RAW SOCKET RACKET SNIFFER

Packet captured!

Source IP : 192.168.1.10

Destination IP : 192.168.1.1

Protocol : TCP

Packet length : 60 bytes

Packet captured!

Source IP : 192.168.1.1

Destination IP : 192.168.1.10

Protocol : ICMP

Packet length : 84 bytes

RAW SOCKET

PACKET SNIFFING

Exp no: 14

Aim:

To implement packet sniffing using RAW sockets and analyze the packets received through the network interface.

Procedure:

1. Create a RAW socket using the socket programming interface.
2. Select the required network interface
3. Enable packet reception through the RAW socket.
4. Capture incoming network packets.
5. Read the packet header.
6. Extract important fields such as:
   - Source IP
   - Destination IP
   - Protocol
   - Packet length
7. Identify the protocol of the captured packet.
8. Display the packet information
9. Continue capturing packets until the program is terminated.

10/10

Result:

Thus, RAW sockets were successfully used to capture and analyze network packets.

Sample output

Webalizer 'server statistics

Total visits : 1250

Total Requests : 6840

Total Pages : 4320

Total Data : 125.6 MB

Top Requested Page:

/index.html

Top browser:

Chrome

Top Operating System:

Windows

WEBALIZER

Exp no: 15

Aim

To analyze different types of servers and their web traffic statistics using the Webalizer tool.

Procedure:

1. Install / configure the webalizer tool.
2. Locate the Web server access log file.
3. Configure Webalizer to read the server log.
4. Run Webalizer on the log file.
5. Allow webalizer to process the access records.
6. Open the generated HTML report.
7. Analyze.
   - Number of visits
   - Number of pages
   - Number of requests
   - Data transferred
   - Client information
   - Referring sites
   - Browser information
   - Operating system information
8. Compare the statistics of different server logs if available.

Result:

Thus, Webalizer was successfully used to analyze web server log files and generate statistical reports about server activity.